World Health Organization

Middle East respiratory syndrome: global summary and assessment of risk

Key highlights:

- The **risk of MERS-CoV transmission** remains **moderate** at both the regional and global levels, reflecting ongoing but restricted patterns of circulation
- Following an abrupt decline in reported MERS human cases in 2020, **notifications have continued in a sporadic and geographically restricted manner**, predominantly within the Middle East region
- WHO expects to see additional cases reported from the Middle East region, with the possibility that cases may be exported to and detected in other countries, as recently happened in France
- **Three limited healthcare-related clusters have been reported from Saudi Arabia** in 2024 and 2025; previous to that, the last cluster had been observed in May 2020, also in Saudi Arabia
- **Three travel-related cases with exposure in the Middle East** have been reported; two were diagnosed in France in November 2025, returning from a trip to the Arabian Peninsula, and one travelled from Saudi Arabia to Pakistan in August 2024, before receiving the MERS diagnosis. International contact tracing revealed no secondary cases
- **Zoonotic spillover of MERS-CoV from dromedary camels continues to represent a primary source of human infection**, typically resulting in isolated human cases without sustained transmission
- **MERS-CoV is enzootic in most dromedary camel populations, within and beyond the Middle East** (e.g. East and North-West Africa, Central and South Asia) and it is important to maintain awareness of possible zoonotic transmission of MERS-CoV in camel-herding regions
- **Recent identification of clade B MERS-CoV in camels from Egypt and Sudan** emphasizes the critical need for ongoing animal surveillance to deepen our understanding of evolutionary patterns and the zoonotic risk posed by emerging strains
- WHO, FAO, WOAH, the Saudi PHA and MoH held a **Quadripartite Global Technical Meeting on MERS-CoV and Other Emerging Zoonotic Coronaviruses** in Riyadh, Saudi Arabia, 27-29 November 2023
- **WHO's MERS case line list has been updated** following a detailed case data review shared by the Saudi MoH and updated case information received from Jordan, Kuwait, Oman, Qatar, United Arab Emirates
- **WHO's MERS dashboard went live** in June 2025 [https://data.who.int/dashboards/mers]

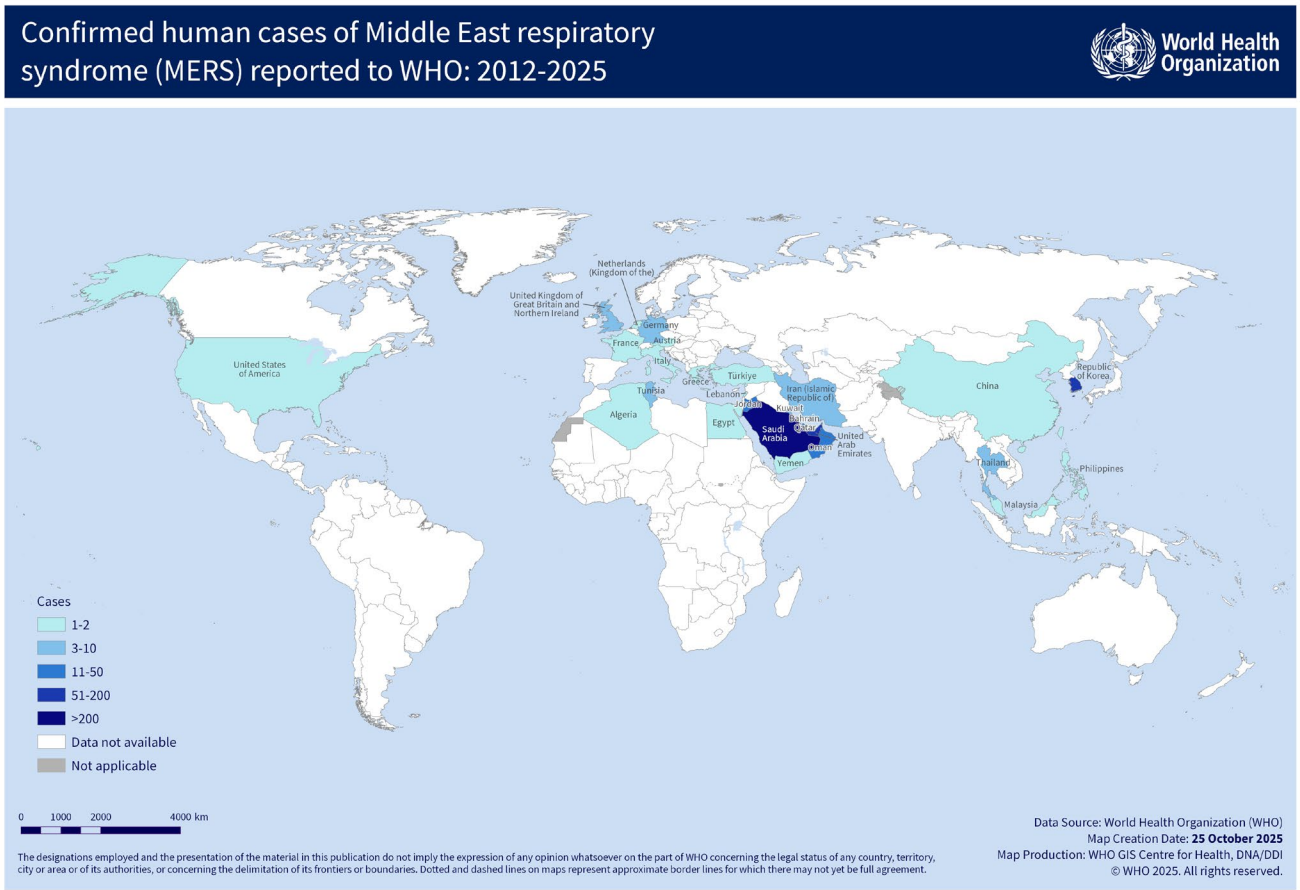
Summary

Since 2012 and as of **5 January 2026**, 2635 laboratory-confirmed cases of Middle East respiratory syndrome coronavirus (MERS-CoV) infection have been reported to WHO from 27 countries in the Middle East, North Africa, Europe, the United States of America, and Asia (Figure 1)[1]. The majority of all cases (84%; 2224/2635) have been reported by Saudi Arabia.

To date, 964 MERS deaths have been reported to WHO, implying a crude case fatality ratio of 37% (95% CI 34.7% - 38.4%).

[1] Numbers of cases, deaths and ratios presented in this document are based on data reported to WHO through the International Health Regulations (2005) or by Ministries of Health, unless otherwise noted.

Figure 1: Confirmed human cases of Middle East respiratory syndrome (MERS) reported to WHO, 2012-2025



Recent MERS line list updates

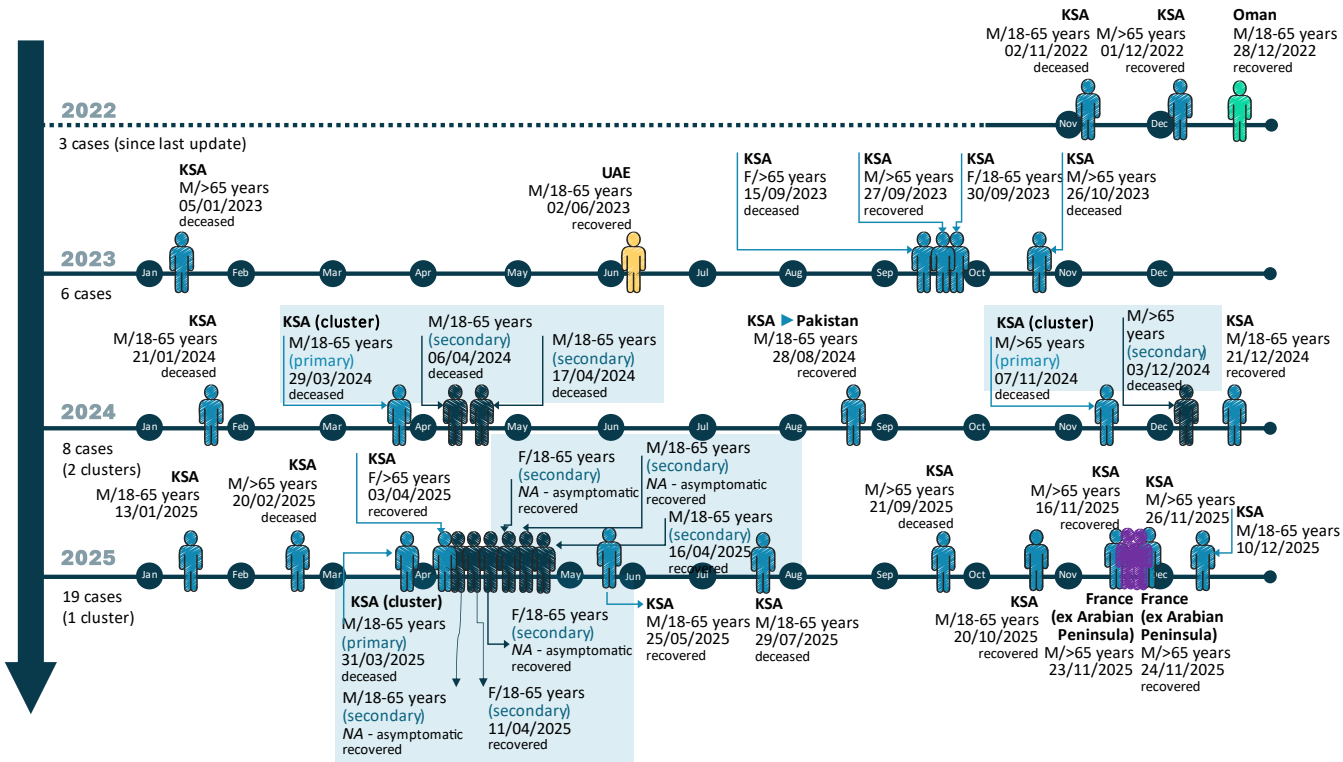
Following the COVID-19 pandemic, the Saudi Ministry of Health conducted a comprehensive retrospective audit of approximately 2200 previously reported MERS cases (up to August 2024). This review required retrieving and re-examining individual hospital records dating back to 2013 to ensure that all data were accurate, complete and internally consistent. As part of this effort, several entries were corrected and standardized, including patient age, date of hospitalisation, and clinical outcomes. The updated dataset was shared with WHO to facilitate revision of Saudi Arabia's entries in the global MERS line list. During this process, it was discovered that 4 cases (two from 2013, one from 2014 and one from 2017) had not been reported via the International Health Regulations (IHR). These cases have now been included within the WHO line list, resulting in a total of 2221 MERS cases reported by Saudi Arabia to date, of which 867 (39%) died. Most epidemiological variables showed only minor adjustments. However, notable refinements were observed in case-type classification and hospitalization reporting, alongside enhanced detail on clinical symptoms, healthcare worker status, camel exposure, and underlying health conditions. Revisions to symptom onset dates also produced subtle shifts in the epidemic curve, including slight increases or reductions in the magnitude of certain outbreak peaks.

In addition, WHO obtained updated case information from Jordan, Kuwait, Oman, Qatar and the United Arab Emirates, further strengthening the accuracy of regional surveillance data (see below).

Description and analysis of recent MERS cases reported to WHO

Since the last global update published on 16 November 2022 and as of 5 January 2026, 36 laboratory-confirmed cases of MERS have been reported to WHO from four countries (32 from Saudi Arabia, 2 from France, 1 from the United Arab Emirates, and 1 from Oman; Figure 2), of whom 14 cases (39%), all from Saudi Arabia, have died. Among these 36 cases, 30 (83%) were male and 6 (17%) were female. The median age was 60 years (IQR 48–73). In addition, four MERS cases from 2013 (n=2), 2014 (n=1) and 2017 (n=1), including 3 deaths, were retrospectively confirmed to WHO from Saudi Arabia in connection with their line list update as well as updated case outcomes from Oman (1 death) and Qatar (2 deaths) (see Table 3 for details).

Figure 2: Timeline of confirmed human cases of Middle East respiratory syndrome (MERS) reported to WHO since the last update (16 November 2022 – 5 January 2026). Case descriptions include reporting country, sex, age range, date of symptom onset and outcome (if available).



Since the last update, around 75% (27/36) of cases reported at least one underlying condition, including chronic renal failure, heart disease, diabetes mellitus, and hypertension. Regarding asymptomatic or mild cases, the proportion has decreased from 22% in 2019 to 19%. This is likely influenced by a revision of the regulations around contact testing in Saudi Arabia in May 2018, now implying that only symptomatic contacts of MERS cases are being tested for the virus. [1] Recently Saudi Arabia reported three health-care-related clusters, two in 2024 comprised of three and two cases each, and one in 2025 comprised of 7 cases (see details further below). Extensive contract tracing was applied in the 2025 cluster, exceeding national regulatory requirements, which lead to detection of four asymptomatic and two mild cases. They fully recovered in home isolation, with the exception of a mild case who was admitted to hospital until recovery as a precautionary measure because he had underlying health conditions. Despite these recent clusters, zoonotic virus transmission remains an important mode of human infection, leading to isolated cases and limited onwards transmission between humans.

Two traveller cases were notified to WHO during the reporting period, with exposure in the Middle East. The two men over 65 years old were diagnosed in France in November 2025, returning from a trip to the Arabian Peninsula where they reportedly had direct and indirect contact with camels. Genomic sequencing was conducted from the first case and reported as being the same lineage that is circulating in the Arabian Peninsula. The other co-travellers remained healthy and international contact tracing revealed no secondary infections. In addition, one case in 2024 travelled to his home country before his diagnosis was known. The person had been admitted to a hospital in Saudi Arabia as a cardiac case where he was also tested for infectious respiratory pathogens as part of national sentinel surveillance. The case decided to discharge himself from the hospital and travel to Pakistan, their home country, to receive care there. Once the MERS diagnosis was known, Saudi health officials immediately contacted their counterparts in Pakistan where the case was put in isolation. Contact tracing in both countries revealed no secondary infections.

Reduction in number of MERS cases reported to WHO since 2020

The number of MERS cases reported to WHO has substantially declined since the beginning of the COVID-19 pandemic in 2020 (Figure 3). Across 2016 - 2019, there were a total of 879 cases reported to WHO (average of 220 cases per year). Since 2020, there have been a total of 124 cases reported (average of 21 cases per year). A detailed review and analysis of viral characteristics pertaining to recently circulating MERS-CoV strains [2] has been published by WHO's Technical Advisory Group for Virus Evolution (TAG-VE) and several hypotheses for the decrease in human case reports have been brought forward:

Role of non-pharmaceutical interventions and behaviours in reduced human-to-human transmission and / or zoonotic spillover incidence: During the early years of the COVID-19 pandemic, public health and social measures (PHSMs; e.g. mask wearing, hand hygiene, physical distancing, improving ventilation of indoor spaces, respiratory etiquette, stay-at-home orders, reduced mobility) likely contributed to a reduction in the transmission of various respiratory pathogens. Although MERS-CoV does not manifest in sustained community transmission, enhanced infection prevention and control (IPC) practices in healthcare settings may have reduced nosocomial spread, and subtle behavioural changes at the camel-human interface may have indirectly decreased zoonotic spillover events, though this remains unproven due to lack of evidence. This may explain why no nosocomial cases were reported in the period 2020-2024. However, as these measures have largely been lifted in recent years, they cannot fully explain the continued reduced level of reported MERS cases compared to the 2012–2019 period.

Reduced surveillance: While MERS-CoV surveillance might have been compromised during the emergency phase of the COVID-19 pandemic, this is now unlikely as since 2023 testing volumes remain high in countries with the highest incidence, including through sentinel surveillance systems and the use of sensitive case definitions for identifying suspected cases. E.g. over 300,000 MERS-CoV tests were conducted in Saudi Arabia between 2023 and 2025, with a positivity rate ranging from 0.03% to 0.05% [3].

Cross-protection conferred by SARS-CoV-2 infection and / or COVID-19 vaccination: Preliminary observational cohort studies suggest that individuals with prior SARS-CoV-2 exposure, while lacking cross-neutralizing antibodies, exhibit cross-reactive antibody responses to spike S2 domain and T-cell responses to MERS-CoV [4,5]. This warrants further investigation.

Reduced virulence of recently circulating MERS-CoV strains: This possibility remains under investigation; however, preliminary data made available to TAG-VE for analysis do not support significant attenuation compared to earlier strains. Conclusive evidence requires further genomic and phenotypic characterization.

Reduced circulation in camels: This is unlikely, as recent studies confirm continued high rates of infection in camels [6] and 36% (8/22; Table 1a) of primary cases reported since the last update mentioned exposure to camels or their products as a possible source of infection.

Human-to-human transmission of MERS-CoV

To date, evidence from both molecular and serologic testing of reported cases indicates that human-to-human transmission of MERS-CoV remains limited. The majority of secondary transmission has occurred within healthcare settings, and this continues to be a defining characteristic of MERS-CoV epidemiology. WHO continues to work with health authorities in affected countries to prevent and minimize health-care-associated cases through improving infection, prevention and control measures in health-care facilities in affected areas and encouraging enhanced outbreak preparedness. To date, MERS remains a relatively rare disease for which medical personnel in healthcare facilities, even in the Middle East, generally have low awareness. In addition, as symptoms of MERS-CoV infection are non-specific, ranging from asymptomatic/no symptoms to pneumonia and acute respiratory distress syndrome, initial cases presenting with mild respiratory signs and symptoms are often easily missed or confused for another respiratory illness. Thus, awareness of MERS-CoV must be increased and testing must be included in respiratory surveillance testing algorithms in relevant areas.

The drivers and precise modes of transmission in healthcare environments remain incompletely understood and are the focus of ongoing collaborative scientific investigations in affected countries. From observational studies, transmission in health-care settings is believed to have occurred before cases were recognized and isolated, and before appropriate IPC procedures were fully implemented. Numerous investigations at the time of the nosocomial outbreaks indicate that aerosolizing procedures conducted in crowded emergency departments or medical wards with sub-optimal IPC measures or adequate ventilation measures lead to human-to-human transmission and environmental contamination. Haemodialysis facilities are at particular risk [7] of onwards MERS-CoV transmission, as dialysis patients are easily exposed through needling, transfusion, and dialysis catheters. In addition, dialysis patients are prone to severe infection, which can increase pathogen shedding. With strong adherence to IPC measures and environmental controls, such as ventilation in affected countries, human-to-human transmission in healthcare facilities can and has been effectively prevented, and at minimum reduced.

Recent clusters of MERS-CoV infections

Since the last update, three health-care-associated clusters of MERS-CoV infection have been reported to WHO, all originating from Saudi Arabia (see Figure 2 for details). Prior to that, the last health-care-associated cluster had been reported in May 2020, also in Saudi Arabia and comprised of six cases.

The first cluster, comprised of three cases, occurred in April-May 2024 in a hospital and involved a male index case in the age range 50-65 years with a history of hypertension and chronic renal failure, who passed away. Two secondary cases were identified, both in the same age category and likewise with fatal outcome. The three cases were epidemiologically linked to exposures in a health-care facility in Riyadh. Contact tracing did not identify any additional cases. The index case had no known exposure to camels.

The second cluster, comprised of two cases, was identified in November 2024 in a haemodialysis facility in Riyadh and involved a male index case aged >65 years undergoing regular haemodialysis, who passed away. He developed hypotension, fever, cough, and dyspnoea during a dialysis session and tested positive for MERS-CoV. Among his 42 contacts, one secondary case was identified: a >65-year-old male receiving haemodialysis at the same facility. He subsequently passed away. The index case had no known exposure to camels.

The third cluster, comprised of seven cases, occurred in April 2025 and involved a male index case in the age range 50-65 years with multiple comorbidities. He underwent hip surgery at a private hospital at the end of February 2025, developed respiratory symptoms while recovering, and tested positive for rhinovirus. He was discharged at the end of March 2025, but was readmitted within hours to another facility in Riyadh due to respiratory distress, where he tested positive for MERS-CoV and eventually passed away in April 2025. As a result of extensive contact tracing in the second facility and testing of all contacts, six healthcare workers (two physicians, three nurses, and one cleaner) tested positive for MERS-CoV: one nurse and one physician developed mild symptoms, while the rest remained asymptomatic. All household contacts tested negative. Over 400 additional contacts were identified and all tested negative. The index case had no known exposure to camels.

Contract tracing during the April 2025 cluster included testing of all contacts, thereby exceeding current regulations [1] in Saudi Arabia which require testing only for contacts developing suggestive MERS symptoms, asymptomatic contacts who had intense contact with a confirmed case, have chronic underlying medical conditions, or are immunocompromised. In doing so, four

asymptomatic contacts were identified which otherwise would not have been reported. Therefore, the relatively high number of secondary case reports (n=6) in this cluster needs to be contextualized accordingly.

Zoonotic transmission of MERS-CoV

Improvements in multi-sectoral investigations of community-acquired MERS cases are evident, including testing of dromedary camels/ herds in the vicinity of community-acquired laboratory-confirmed cases reporting direct or indirect contact with camels or their products through occupation, recreation, husbandry, or other. Typically, Ministries of Health in affected countries notify the Ministries of Agriculture when human cases report an exposure or potential link to dromedary camels or their products.

In recent years zoonotic virus transmission has been an important source of human MERS-CoV infection, generally leading to isolated human cases. During the reporting period, 41% (11/27) of primary cases, i.e. those without known exposure to a human MERS case, mentioned direct or indirect exposure to camels or their products prior to disease onset (Table 1a).

Of note is that, while Africa hosts about 80% of the global camel population, no locally acquired zoonotic human case has been officially reported to date. This may be in part explained by a lack of targeted surveillance activities and testing, even though research studies provided evidence of human exposure. Studies comparing virus isolates from dromedary camels found that clade C virus (circulating in Africa) viral replication levels and transmission efficiency is lower in camelids than clade B virus (circulating in the Middle East) [8]. For the first time, clade B viruses have now been identified in camels of African origin [9,10], highlighting the risk of introduction from the Arabian Peninsula into African camel populations and further spread in light of its increased replication competence and more efficient viral entry over Clade C. [11]

Table 1a. Number of cases reportedly exposed to camels / camel products since last risk assessment (16 November 2022 – 5 January 2026)

Timeframe	Nr exposed to camels	Nr primary cases	Percentage (of primary)	Nr total cases	Percentage (of total)
16 November 2022 – 5 January 2026	11	27	41%	36	31%

Table 1b. Number of cases reportedly exposed to camels / camel products each year since 2020

Year	Nr exposed	Nr primary cases	Percentage (of primary)	Nr total cases	Percentage (of total)
2020	10	49	20%	60	17%
2021	12	20	60%	20	60%
2022	6	7	86%	9	67%
2023	3	5	60%	6	50%
2024	2	5	40%	8	25%
2025	5	13	38%	19	26%
Total	**38**	**99**	**38%**	**122**	**31%**

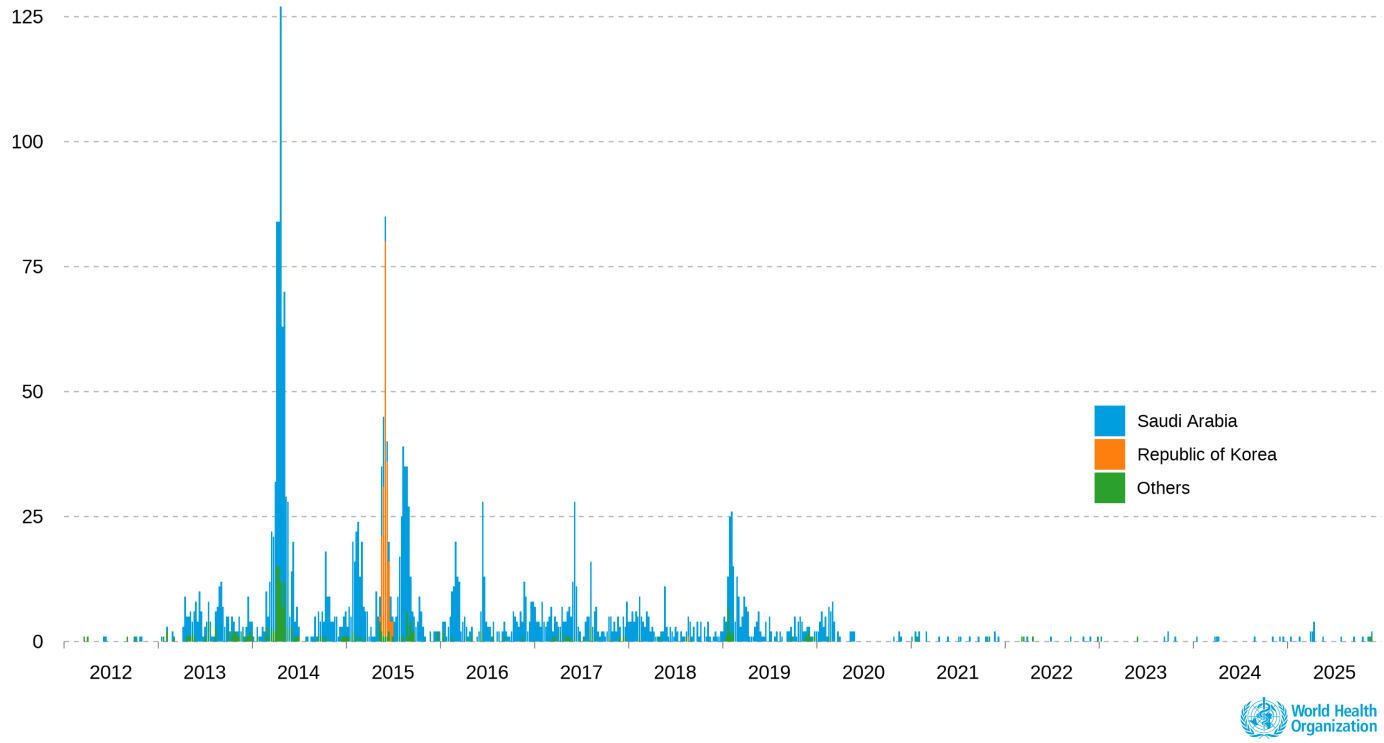
Table 2. Cases of MERS-CoV infection with indication of zoonotic exposure reported to WHO since last risk assessment (16 November 2022 – 5 January 2026)

Country	Sex	Age group	Underlying condition	Date of symptom onset	Symptoms	Animal contact in the 14 days prior to symptom onset	Outcome	Nr of contacts testing positive (in %)
Saudi Arabia	M	>65	diabetes mellitus benign prostatic hyperplasia	01.12.2022	fatigue, cough, runny nose, sore throat shortness of breath	direct contact with camels, consumption of their raw milk	recovered	0/8 (0%)
Saudi Arabia	M	>65	diabetes mellitus hypertension	05.01.2023	fever, cough, shortness of breath	direct contact with camels, consumption of their raw milk	deceased	0/7 (0%)
Saudi Arabia	M	>65	hypertension diabetes mellitus dyslipidemia chronic obstructive pulmonary disease	27.09.2023	fever, cough, headache, sore throat, shortness of breath	indirect contact with camels (had contact with his brother who owns camels)	recovered	0/19 (0%)
Saudi Arabia	M	>65	bronchial asthma	26.10.2023	fever, cough, headache, shortness of breath	direct contact with camels (camel owner)	NR	0/50 (0%)
Saudi Arabia	M	18-65	hypertension, diabetes mellitus, history of renal transplantation	21.01.2024	fever, cough, shortness of breath, rhinorrhea, fatigue	direct contact with camels (frequent visits to camel market)	deceased	0/48 (0%)
Saudi Arabia	M	18-65	diabetes mellitus hypertension end-stage renal disease on haemodialysis	21.12.2024	fever, cough, shortness of breath, vomiting	consumption of raw camel milk	NR	0/55 (0%)
Saudi Arabia	M	>65	diabetes mellitus end-stage renal disease on haemodialysis	20.02.2025	fever, cough, shortness of breath	indirect camel contact	deceased	0/45 (0%)
Saudi Arabia	M	18-65	chronic kidney disease colon cancer managed with surgery and chemotherapy	25.05.2025	fever, cough, shortness of breath	indirect camel contact	NR	0/21 (0%)
France	M	>65	NR	23.11.2025	mild respiratory symptoms for 1 day	direct and indirect camel contact in the Arabian Peninsula	recovered	0/60 (0%)
France	M	>65	several	24.11.2025	pneumonia	direct and indirect camel contact in the Arabian Peninsula	recovered	0/60 (0%0
Saudi Arabia	M	>65	diabetes mellitus, hypertension, dyslipidemia, benign prostatic hyperplasia, nephropathy post kidney transplant	26.11.2025	shortness of breath tachypnea	direct camel contact	NR	0/38 (0%)

Nine out of the eleven cases with potential direct or indirect zoonotic exposure were from Saudi Arabia and all were male, 73% (8/11) were older than 65 years, 27% (3/11) died and for ten out of the eleven cases underlying medical conditions were reported. Onset dates were between September and May. None of their overall 411 contacts developed symptoms.

Global summary of MERS cases since 2012

Figure 3. Confirmed MERS cases reported to WHO since 2012 and as of 5 January 2026 (by week; N=2635; source: WHO MERS dashboard) [12]



At the time of publishing, the date of symptom onset was not available for 28 cases.

Symptomatic cases are plotted by date of symptom onset; asymptomatic cases are plotted by date of notification to WHO.

blue = Kingdom of Saudi Arabia; orange= Republic of Korea; green = all other countries reporting MERS cases to date including Algeria, Austria, Bahrain, China, Egypt, France, Germany, Greece, Iran, Italy, Jordan, Kuwait, Lebanon, Malaysia, the Netherlands, Oman, the Philippines, Qatar, Thailand, Tunisia, Türkiye, United Arab Emirates, the United Kingdom, the United States, Yemen.

Of the 2635 laboratory-confirmed cases reported to WHO since 2012 (Figure 3), the median age is 53 years (IQR 38-66 years) and 1834 (70%) are male. 56% (1486/2635) cases reported to WHO also reported having at least one of the following underlying conditions: diabetes mellitus, hypertension, heart disease, chronic renal failure, or lung disease. 77% (743/964) of fatal MERS cases had at least one of these underlying conditions. The crude case fatality ratio is 37% (964/2635), increasing to 50% (743/1486) for cases with comorbidities and 67% (580/869) for MERS patients admitted to ICU.

Since 2012, 27 countries from all WHO regions have reported human cases of MERS-CoV infection. In the Middle East: Bahrain, Egypt, Iran (Islamic Republic of), Jordan, Kuwait, Lebanon, Oman, Qatar, Saudi Arabia, the United Arab Emirates and Yemen; in Africa: Algeria and Tunisia; in Europe: Austria, France, Germany, Greece, Italy, the Netherlands (Kingdom of the), Türkiye and the United Kingdom of Great Britain and Northern Ireland; in Asia: China, Malaysia, the Philippines, the Republic of Korea and Thailand; and in the Americas: the United States of America (Table 3). The majority of cases (2224/2635, 84%) have been reported from Saudi Arabia (Table 3).

Table 3. Number of laboratory-confirmed MERS cases reported by countries, since 2012 and as of 5 January 2026.
Information updated since 16 November 2022 is highlighted in red.

Country reporting	Number of cases	Number of deaths	Case Fatality Ratio (%)	Number of cases since last update (16.11.2022)	Onset date of last case (DD/MM/YYYY)
Algeria	2	1	50%	0	23/5/2014
Austria	2	1	50%	0	22/9/2014
Bahrain	1	1	100%	0	4/4/2016
China	1	0	0%	0	19/5/2015
Egypt	1	0	0%	0	22/4/2014
France	4	1	25%	2	24/11/2025
Germany	3	2	67%	0	10/2/2015
Greece	1	1	100%	0	8/4/2014
Iran (Islamic Republic of)	6	2	33%	0	18/3/2015
Italy	1	0	0%	0	24/5/2013
Jordan	28	12	43%	0	26/9/2015
Kuwait	4	2	50%	0	8/9/2015
Lebanon	2	0	0%	0	8/6/2017
Malaysia	2	1	50%	0	14/12/2017
Netherlands (Kingdom of the)	2	0	0%	0	5/5/2014
Oman[1]	26	8	31%	1	28/12/2022
Philippines	2	0	0%	0	30/6/2015
Qatar[1]	28	10	36%	0	11/3/2022
Republic of Korea	186	36	19%	0	26/8/2018
Saudi Arabia[1]	2224[2]	869	39%	32	10/12/2025
Thailand	3	0	0%	0	28/7/2016
Tunisia	3	1	33%	0	13/5/2013
Türkiye	1	1	100%	0	25/9/2014
United Arab Emirates	94	12	13%	1	2/6/2023
United Kingdom of Great Britain and Northern Ireland	5	3	60%	0	13/8/2018
United States of America	2	0	0%	0	1/5/2014
Yemen	1	1	100%	0	17/3/2014
Total	2635	964	37%	36	10/12/2025

[1] Death counts for these countries were updated according to additional information received from Member States
[2] Includes 4 cases from 2013 (n=2), 2014 (n=1) and 2017 (n=1) that have been retrospectively reported to WHO by the Saudi Ministry of Health

Based on available data, populations in close contact with dromedary camels (e.g. farmers, abattoir workers, shepherds, dromedary owners) and health and care workers caring for MERS patients are at higher risk of MERS-CoV infection (Spasenoska, 2025 – manuscript submitted for publication). Healthcare-associated transmission has been documented in several countries, including Saudi Arabia, Jordan, the United Arab Emirates, France, the United Kingdom, and the Republic of Korea with varying outbreak sizes (2-180 reported cases per outbreak). Among the 2635 MERS cases reported to WHO to date, at least 17% (448/2635) have been health and care workers.

The largest outbreak outside of the Middle East occurred in the Republic of Korea in 2015 resulting in 186 cases (including one case who travelled to China) and 39 deaths. An additional exported case to the Republic of Korea in September 2018 remained an isolated event and did not result in further human-to-human transmission. This was likely attributable to early suspicion of MERS-CoV infection, immediate isolation of the case from others, appropriate clinical management of the patient upon his return from Kuwait, and thorough contact tracing to ensure all contacts were captured and screened for symptoms.

How work on MERS-CoV informed the early COVID-19 response

The WHO Global Program of Work for MERS-CoV and its Public Health Research Agenda successfully informed the early response to COVID-19, serving as a crucial foundation for actions by Member States and WHO during the pandemic.
For example:

- Expert networks for MERS-CoV including laboratory, IPC, mathematical modelling and clinical care networks were rapidly utilized for COVID-19;
- Reference laboratories for MERS-CoV were leveraged, particularly in the early stages of the pandemic, to provide laboratory support and confirmatory testing for affected countries – this network has now been institutionalized as WHO Coronavirus Network (CoViNet) [13];
- Technical guidance documents, training materials, commodity packages, assessment tools and checklists developed for MERS-CoV were used to inform initial technical guidance documents and information products for COVID-19;
- Standardized sero-epidemiological protocols developed for MERS-CoV were quickly adapted for COVID-19 and implemented across many countries to rapidly understand the extent of infection, risk factors for infection, and to identify key epidemiologic parameters of SARS-CoV-2;
- The ChAdOx1 vaccine platform technology developed for MERS-CoV was utilized to develop the Oxford/AstraZeneca COVID-19 vaccine.

Recent WHO MERS activities

The following MERS-related activities were conducted by WHO since the publication of the last global update:

- Since its designation in January 2021, the WHO Collaborating Centre for MERS, the Public Health Authority (PHA) in Saudi Arabia [14], has actively supported WHO's Global Programme of Work on MERS-CoV by contributing to the development of technical guidance, conducting collaborative research on MERS-CoV, assisting in the nomination of experts for WHO guidelines, carrying out systematic reviews, and supporting the implementation of Unity Study investigation reports and related laboratory activities.
- A WHO EPI-WIN webinar [15] organized on 24 May 2023 discussed why global work on MERS-CoV prevention, surveillance and control remain important for pandemic preparedness and prevention, what has been learnt from MERS and what gaps still remain.
- During the 2023 EMARIS Conference held in Muscat, Oman, WHO together with the Food and Agriculture Organization of the United Nations (FAO) and the World Organisation for Animal Health (WOAH) conducted a multi-country MERS-CoV simulation exercise to test and enhance regional capacities in risk assessment, multi-sectoral coordination, and response to a potential cross-border outbreak scenario.
- The FAO-WHO-WOAH-United Nations Environment Programme (UNEP) Quadripartite, in partnership with the PHA and the Ministry of Health, Saudi Arabia, held a Quadripartite Global Technical Meeting on MERS-CoV and Other Emerging Zoonotic Coronaviruses in Riyadh on 27-29 November 2023. The meeting was attended by public health officials, researchers and resource partners and industrial partners across the global community. The aims were to share the latest efforts from implementing the MERS-CoV public health research agenda and research and development roadmap, with a strong emphasis on a One Health approach. At the end of the meeting participants agreed on priority research gaps and engaged in a marketplace activity designed to identify and kickstart multi-disciplinary collaborations to start addressing these gaps. Details can be found in the meeting report [16].
- In November 2023, WHO updated its Managing epidemics handbook [17], providing concise and basic up-to-date knowledge with which public health officials can respond effectively and rapidly at the very start of an outbreak. MERS is one of the 19 deadly diseases included, with key facts and tips for outbreak response interventions.
- The EMRO-developed operational Framework for integrated respiratory disease surveillance [18], prioritizing MERS-CoV alongside other high-threat respiratory viruses within existing influenza sentinel surveillance systems, is actively being implemented in select countries, including Saudi Arabia and Qatar. Implementation is planned for Somalia and Pakistan, with preparations underway in coordination with national authorities. Additionally, WHO's Regional Office for Africa (AFRO) is reviewing the applicability of this framework for countries with large camel populations and trade routes to support targeted MERS-CoV surveillance and early detection efforts.
- Following a meeting in December 2023 [19], the Global Influenza Surveillance and Response System (GISRS) has been expanded to also collect surveillance data for other respiratory viruses such as SARS-CoV-2, Respiratory Syncytial Virus (RSV) and MERS-CoV. e-GISRS monitors the co-circulation of respiratory viruses and provides training, technical, and logistical support to resource-limited countries to conduct testing.
- A working group between the technical MERS Team and the Community Protection and Resilience Unit, including the Risk Communication and Community Engagement (RCCE) team, and the Behavioural Insights (BI) team at WHO HQ and EMRO has been set up in March 2024 to meet every 6 weeks to discuss strategic and technically sound approaches to strengthening community protection and resilience outcomes. This includes engagement and communication with affected communities, identifying and assessing spillover risks at the animal-human interface, as well as co-developing risk mitigation measures with communities. In September 2024, the BI Team undertook a rapid scoping review to identify human-camel interaction behaviours involved in MERS-CoV transmission between camels and humans (unpublished). The review identified a relatively small and heterogenous number of studies and highlighted several gaps in the precise characterization in literature of human-camel interaction behaviours. It identified opportunities for future research to address, such as (1) no studies applied behavioural science theories, frameworks or models to enhance the understanding of MERS-relevant specific human-camel interaction behaviours; (2) relevant behaviours were often not specifically or sufficiently defined or described, nor were sample sizes adequate to detect trends or draw conclusions about behaviour risk; and (3) studies often failed to identify or discuss in sufficient detail the barriers or enablers to relevant behaviours. Focused mixed-methods studies, including observations, are needed to better define behaviours related to camel-human interactions and which help diagnose potential barriers and enablers of behaviour change to reduce risk of MERS-CoV spillover events.
- On 27 March 2024, WHO officially launched the Coronavirus Network (CoViNet, [13]), bringing together laboratories from the human, animal, and environmental health sectors to support integrated surveillance, enhanced virological assessment, and risk assessment of SARS-CoV-2, MERS-CoV and novel coronaviruses of public health importance. The network aims to strengthen early detection, inform public health responses, build capacity and support pandemic preparedness efforts under a One Health approach. The first CoViNet newsletter [20] was published on 24 March 2025. An External Quality Assessment Programme (EQAP) coordinated jointly by the WHO's Global Influenza Programme and CoViNet has incorporated SARS-CoV-2 and MERS-CoV, as targets to be tested by GISRS labs in 2025 to evaluate their performance.
- In May 2024, WHO conducted a regional workshop in Jordan under the Mosaic [21] (Multi-pathogen Optimization of Surveillance, Analytics, and Infrastructure Coordination) initiative. The workshop convened stakeholders from national public health institutions, influenza surveillance networks, and technical partners to advance implementation of an integrated

respiratory surveillance framework. Emphasis was placed on leveraging existing systems such as e-GISRS to incorporate multiple respiratory pathogens, including MERS-CoV, into routine surveillance. This activity aimed to enhance regional capacities for early detection, coordinated response, and evidence-based risk assessment of respiratory threats with epidemic and pandemic potential.

- In August 2024, WHO officials from headquarters and EMRO, and the WHO collaborating centre for infectious disease modelling at Imperial College London, conducted a technical mission to the Kingdom of Saudi Arabia to meet with senior officials from the Ministry of Health (MOH) and the Public Health Authority (PHA) responsible for MERS-CoV surveillance, case management, IPC, data collection and analysis, database management and training. The aims of the mission were to synchronize the MERS database, discuss a minimum dataset for MERS case reporting under the IHR and enhance surveillance and response coordination. After the COVID-19 pandemic, the Saudi MoH undertook a retrospective review of their approximately 2,200 reported MERS cases (to August 2024). This involved going back to individual hospital records from as early as 2013 to ensure that data were up-to-date and consistent, which included updating entries which were incorrect (e.g. age, hospitalisation date, outcome). The updated data were shared so that WHO could update the Saudi data in the global line list accordingly.
- In September 2024, the WHO in collaboration with the PHA conducted a National Simulation Exercise Workshop on Multisectoral Coordination for MERS-CoV Preparedness, Prevention, and Control in Riyadh, Saudi Arabia. The exercise aimed to strengthen preparedness, enhance coordination among sectors, and refine strategies for tackling zoonotic diseases like MERS. The workshop highlighted the necessity of multisectoral coordination in addressing zoonotic diseases such as MERS. Through a collaborative One Health approach, stakeholders across human, animal, and environmental health sectors must work together to strengthen surveillance, enhance laboratory capacity and data sharing, and implement targeted interventions. The key issues evidenced by the simulation exercise are being used by MoH and PHA to review and update national MERS-CoV preparedness and surveillance plans and will serve as a model for other countries confronting similar emerging threats.
- WHO updated its MERS-CoV investigation protocols [22] which were published in 2024 as part of WHO's Respiratory Investigations and Studies (Unity Studies):
    - How to conduct surveillance and investigations of human infection with Middle East respiratory syndrome coronavirus using WHO's Investigations and Studies (Unity Studies 2.0) protocols (see Figure 4)
    - How to investigate the first few X cases and contacts of human infection with Middle East respiratory syndrome coronavirus
    - How to conduct a case-control study to assess the potential risk factors related to human illness caused by Middle East respiratory syndrome coronavirus
    - How to conduct a cohort study to assess the potential risk factors of Middle East respiratory syndrome coronavirus infection among health and care workers in a health-care setting
    - How to sample surfaces in health-care settings for Middle East respiratory syndrome coronavirus
    - How to conduct a cross-sectional study of Middle East respiratory syndrome coronavirus in populations occupationally exposed to dromedary camels

  The protocols and their use were introduced during a WHO EPI-WIN webinar [23]. Arabic and French translations of the protocols are in preparation.
- In November 2024 WHO launched the MERS Tracker dashboard [33] in collaboration with the University of Calgary SeroTracker team and FAO. The dashboard provides users with centralised access to serological and virological MERS-CoV field studies (Figure 5). Results from these studies can fill important gaps in our knowledge of MERS-CoV distribution: officially reported MERS-CoV infections in humans and camels, predominantly from the Middle East, represent the 'tip of the iceberg'; from field research studies we know that the geographic distribution of MERS-CoV reaches as far as Northern Africa and Central and South Asia (Figure 6). Data extraction and analysis features of the dashboard have been designed to make information and results from these studies accessible for evidence-based decision making by public health stakeholders. A WHO EPI-WIN webinar [24] provided an introduction to the MERS Tracker and how the dashboard can display data according to user preference.
- WHO continues supporting field-based epidemiological studies at the animal-human interface to evaluate the extent of MERS-CoV infection in individuals with occupational exposure to dromedary camels. In Pakistan, a serological survey conducted in January 2020 and March 2021 among camel handlers and their families in two camel-rearing districts of Punjab found evidence of past zoonotic exposure in the absence of reported clinical cases. In late 2024, a follow-up study was launched in Pakistan under the WHO CoViNet initiative to monitor active circulation of MERS-CoV in dromedary camels, aiming to identify and characterize circulating MERS-CoV strains and strengthen understanding of transmission dynamics at the animal–human interface. In Somalia, a study conducted with WHO support since mid-2019 generated the first ever evidence of zoonotic human MERS-CoV infection in this country. Results from these studies will be published in peer-reviewed journals and highlight the need for continued surveillance and research to understand MERS-CoV dynamics and mitigate potential risks in camel-rearing communities of South Asia and northern Africa.
- In March 2025, WHO, in collaboration with FAO and WOAH, launched the Global MERS Community of Practice (CoP) through the WHO HIVE platform in order to provide a collaborative platform for Member States, technical experts, and partners. The MERS CoP enables regular information exchange, peer-to-peer learning, discussion of emerging evidence, and

alignment of approaches for surveillance, preparedness, and response to MERS-CoV, under the One Health framework. Several technical discussions and webinars have already been held on the platform.

- WHO launched an online MERS-CoV introductory training course [25] in two modules, accessible from OpenWHO.
- On 4 June 2025, WHO launched its first ever MERS-CoV data dashboard [12, https://data.who.int/dashboards/mers] through an EPI-WIN webinar. [26] The dashboard provides public access to WHO's global MERS case data, enabling real-time visualizations of key epidemiological variables, geographic distribution, and trends over time. The dashboard enhances transparency, supports evidence-based decision-making, and facilitates rapid risk assessment and response at global, regional, and national levels.
- WHO has updated its infographics on MERS-CoV, available on the WHO MERS-CoV health topics [27] webpage under 'Infographics', and the fact sheet. [28]
- WHO collaborated with the Pandemic Sciences Institute at the University of Oxford, United Kingdom of Great Britain and Northern Ireland to conduct an evidence review on prevention and mitigation of MERS-CoV spillover and community protection using a rapid research needs appraisal (RRNA) methodology to quickly and robustly identify gaps in knowledge and evidence. The review revealed limited, predominantly observational evidence. Demographic reporting was scarce, and research has largely been restricted to the Arabian Peninsula, with little data from the African and other regions where risk of spillover exists. Further multi-disciplinary research and operational studies are needed to fill existing knowledge gaps. These should include longitudinal occupational cohort studies, as well as stand-alone social and behavioural studies, using qualitative and quantitative methods and including at-risk populations in various settings. Outcomes of the review will inform technical products and tools aimed at prevention and mitigation of MERS-CoV spillover through understanding and addressing social and behavioural drivers of pathogen spillover for community protection. Results are being prepared for publication in a peer-reviewed journal.
- The WHO Technical Advisory Group for Virus Evolution (TAG-VE) conducted a risk evaluation of circulating MERS-CoV clades and lineages [2], published on 22 August 2025. The group reviewed current evidence from global genomic MERS-CoV surveillance to highlight gaps in existing knowledge, propose recommendations for further studies, and evaluate current public health risks associated with this zoonotic virus. The virus continues to circulate at high levels amongst camels and continues to genetically evolve. Recent MERS-CoV isolates from camels in Saudi Arabia demonstrate unique genetic features, including novel amino acid polymorphisms in the spike protein. However, further characterization will be required to understand any phenotypic effects. Reduced virulence of currently circulating MERS-CoV strains, one of the hypotheses brought forward to explain the recent decrease in human case reports, remains under investigation. Preliminary data do not support significant attenuation compared to earlier strains, but conclusive evidence requires further genomic and phenotypic characterization. The risk of MERS-CoV infection in endemic regions is therefore assessed as high for occupationally exposed humans, healthcare personnel, and people with underlying health conditions. More genomic surveillance is needed to assess whether this risk is lower in countries with camel populations outside of the Arabian Peninsula, where genetically different MERS-CoV clades are circulating but a lack of genomic data exists.
- WHO has developed an automated MERS-CoV genomic analysis pipeline that continuously retrieves and curates global sequences, harmonizes metadata to WHO standards, and generates high-resolution phylogenetic and epidemiological summaries across hosts, countries, and years. The analyses provide WHO with near-real-time insights into viral evolution, transmission dynamics, and sampling gaps, supporting evidence-based risk assessment, targeted surveillance, and policy guidance for preparedness and response in affected regions. In parallel, WHO created a dedicated MERS-CoV community dataset for Nextclade [29], which will facilitate standardized global lineage tracking and is planned for integration into the main Nextstrain framework to strengthen public access to MERS-CoV genomic data and situational awareness.
- The WHO BioHub System recently added an isolate of MERS-CoV Clade C. [30] Through the BioHub, countries can voluntarily share and request biological materials with epidemic or pandemic potential. Such rapid access to verified biological materials and data is essential for advancing research, validation, and product development.
- WHO has started the process of updating guidelines on IPC as well as clinical management and diagnosis of MERS. Guideline Development Groups (GDGs) comprised of multi-disciplinary experts have been set up and started meeting on a regular basis. Evidence review teams have been appointed and started identifying existing literature.
- WHO created a Global COVID-19 and other Coronaviruses Programme in January 2025 to consolidate the work on COVID-19, MERS and SARS, and will be including any future emerging coronavirus of public health importance as well. Supporting this programme, and following on from the last Global Strategic Preparedness and Response Plan for COVID-19 (SPRP; 2023-2025), WHO's new strategic and operational plan [31] tackles coronavirus disease threat management comprehensively. Covering the period 2025-2030, the plan sets out the global framework for supporting Member States in the sustained, integrated, evidence-based management of coronavirus disease threats, including MERS, and reaffirms WHO's commitment to strengthen prevention, control and pandemic readiness at global, regional and country levels.

Figure 4: WHO MERS-CoV investigations and studies [22]: when to use which protocol

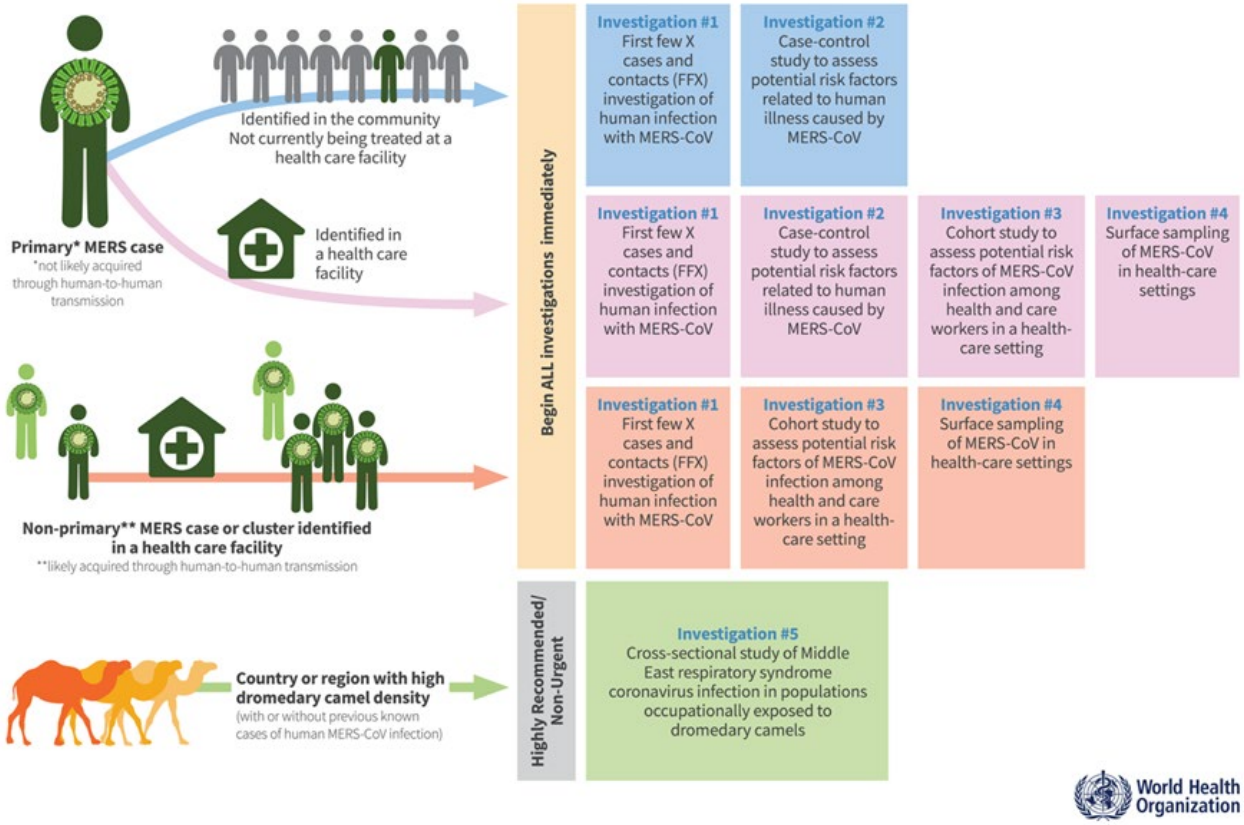


Figure 5. MERS Tracker dashboard [33] displaying number of published MERS-CoV studies in humans and animals in their geographic location, over the camel population density

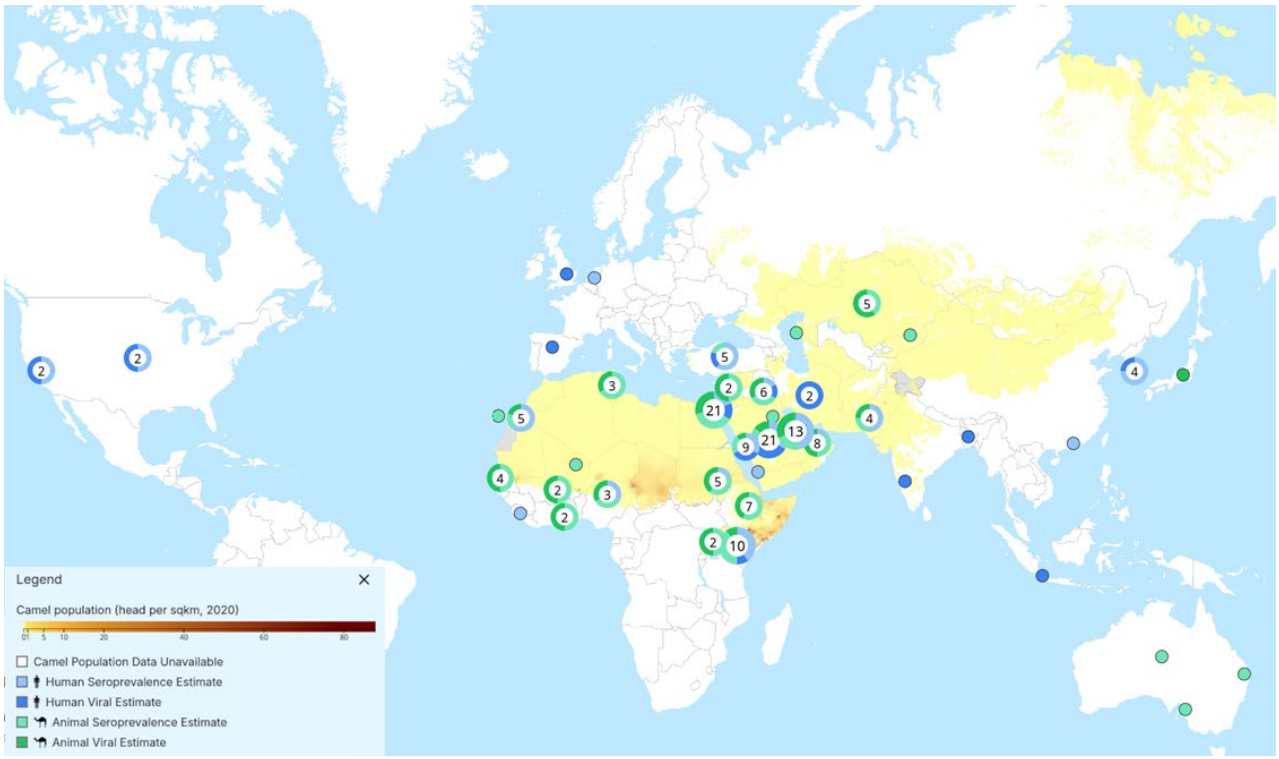
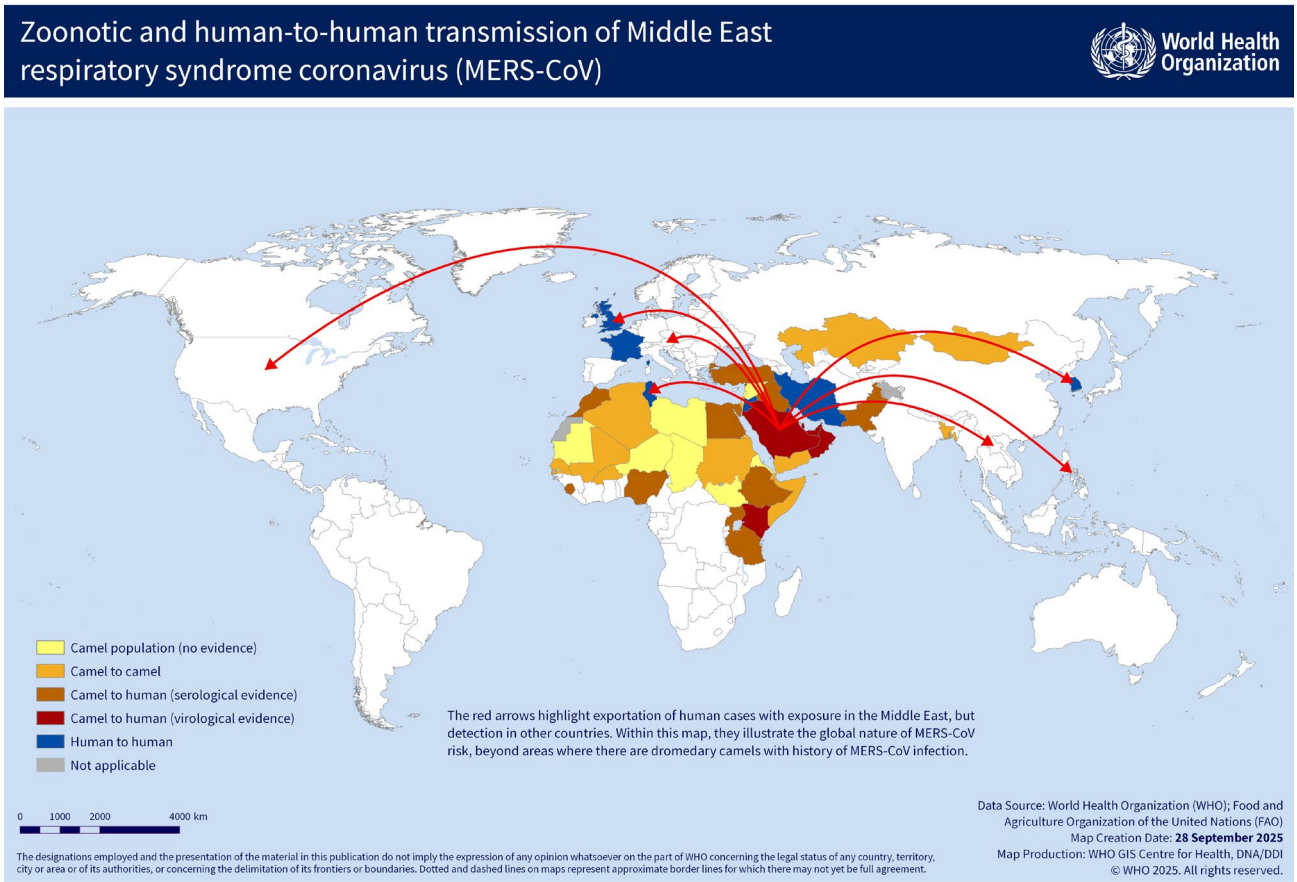


Figure 6. Zoonotic and human-to-human transmission of Middle East respiratory syndrome coronavirus (MERS-CoV)



Risk assessment

The global risk for MERS remains moderate, both at regional and global levels, as assessed by WHO EMRO and HQ.

Persistent circulation: Despite a reduction in human case reporting, in particular since 2020, MERS-CoV continues circulating and genetically evolving in dromedary camels across large parts of the Middle East, northern and Sub-Saharan Africa, and Central and South Asia (Figure 6). [2,6,32,33] For the first time, clade B viruses have been identified in camels of African origin [9,10], highlighting the risk of introduction from the Arabian Peninsula into African camel populations and further spread in light of its potential growth advantage over Clade C. The global MERS threat therefore remains a topic meriting research attention and may not only originate from the Middle East. Although no human MERS cases with locally acquired infection have officially been reported from these other subregions, there is evidence of zoonotic transmission through serology, PCR and T-cell studies in humans [5,34,35] and further investigation and public health awareness of this risk is crucial. The observed decline in reported MERS cases since 2020 (Figure 3), in particular during the COVID-19 pandemic emergency phase, is thought to be a result of pandemic-related IPC measures that also limited human-to-human transmission of MERS-CoV, as well as behavioral changes during the pandemic. Any role of potential cross-reactive immunity from SARS-CoV-2 infection and/ or vaccination remains in need of further investigation. Other hypotheses—such as reduced surveillance, viral attenuation, or decreased circulation in camel populations—are not supported by current evidence.

Continued zoonotic and nosocomial transmission: Since the last update, there have been continued sporadic introductions from dromedary camels into the human population in the Middle East, in particular Saudi Arabia. While secondary human-to-human transmission in health-care settings and households has been noticeably reduced, three small healthcare-related clusters have recently been reported by Saudi Arabia in 2024 and 2025 (Figure 2), that were quickly controlled. Nine nosocomial cases were identified in these clusters, involving fellow patients (3) in the 2024 clusters and health and care workers (6, 4 of which remained asymptomatic) in the 2025 cluster. The large outbreaks in Jeddah/Riyadh, Saudi Arabia in 2014, in the Republic of Korea in June 2015 and in Riyadh, Saudi Arabia in August 2015, serve as a reminder that MERS-CoV, if not adequately controlled, can cause explosive outbreaks with substantial health and socio-economic consequences. If MERS-CoV would lead to an outbreak in a previously unaffected country, as happened in the Republic of Korea in 2015, consequences will be major, not least due to the high crude case fatality rate of 37%.

Surveillance challenges: During the emergency phase of the COVID-19 pandemic (2020-2023), prioritization of epidemiological surveillance activities for COVID-19 and reduced testing for MERS-CoV in affected countries in the Middle East likely resulted in reduced detection of MERS cases. The non-specific respiratory symptoms of MERS, in the context of influenza and COVID-19 co-circulation, further complicate and possibly delay early detection and timely isolation of cases. Existing surveillance systems

focused on acute severe respiratory infections may miss mild cases of MERS. This provides opportunity for onwards human-to-human transmission, especially in health-care settings.

Understanding drivers of transmission: While investigations and studies continue to evaluate transmission within healthcare facilities, more comprehensive scientific studies are needed to better understand the behaviours driving transmission, including investigations into the role of asymptomatic cases and environmental contamination, such as surface and air survival and persistence, as well as the social and behavioural risk factors related to camel raising and other zoonotic risk factors that have not been adequately investigated. Challenges include the accurate tracking of exposures when patients move between hospitals within countries and between countries, e.g. for medical treatments and/or surgery (medical tourism), complicating the epidemiologic picture. Secondary cases have reported varying levels of contact with confirmed patients, ranging from direct contact (e.g., health and care workers providing direct care to infected patients before diagnosis with MERS) to no clear contact (e.g., patients sharing wards with infected patients, but without sharing health and care workers or rooms). Several studies from the Republic of Korea have identified MERS-CoV virus on surfaces inside patient rooms and on equipment during patient stays and after discharge or death. These findings highlight the importance of adequate cleaning and disinfection of patient rooms. Community evidence collected, for example, through rapid assessments for community protection, engagement with community representatives, community dialogue, observations or other operational activities, can provide critical insights into behaviours, lived experiences, barriers to health-seeking, and perceptions of risk and trust. When systematically gathered and integrated into public health decision-making, these outputs can strengthen local ownership of risk-mitigation measures, improve the acceptability and feasibility of IPC measures, and support the design of tailored interventions that are more likely to interrupt transmission and protect those most at risk.

Virus monitoring: Genetic sequencing plays a critical role in understanding transmission and virus evolution and sequencing of samples collected from confirmed patients should be a routine part of investigations into MERS clusters to further our understanding of transmission patterns between patients and to help identify the source of the infection. During a FAO-WHO-WOAH Global technical meeting on MERS-CoV and other emerging zoonotic coronaviruses in 2021 [36], experts suggested that MERS-CoV recombination with SARS-CoV-2, i.e. the creation of a novel virus that combines genetic materials from both viruses when they affect the same host cell at the same time, is a remote possibility. Surveillance systems should be set up so that recombinants and novel variants can be detected in a timely manner, especially in areas where both viruses circulate.

Importance of the One Health approach: Until zoonotic transmission of the virus from infected dromedary camels into the human population is halted, the risk of human infections remains, posing a risk of further spread, particularly in health-care settings. Control of MERS-CoV, as for any other zoonotic coronavirus, requires national leadership, coordination between animal and human sectors (and others) using a One Health approach, public trust, frequent and clear communication to all hospitals and hospital staff on IPC measures to limit human-to-human transmission, thorough investigation of all cases and rapid dissemination of knowledge gained during outbreak investigations and research on MERS-CoV.

Have MERS-CoV transmission patterns changed?

Overall, the epidemiology, transmission patterns and clinical presentation of MERS remained consistent with patterns described in previous WHO global summaries and assessment of risk. That is, MERS-CoV is a zoonotic virus that sporadically enters the human population via direct or indirect contact with infected dromedary camels or camel products, predominantly in the Arabian Peninsula. Human-to-human transmission has occurred in health-care settings and, to a lesser extent, among close community contacts. A recent trend can be observed in that less cases are being reported from endemic countries than in the past, and that the majority of these cases are isolated cases, with or without reported exposure to dromedary camels or their products.

While there is human-to-human transmission detected among some close contacts of confirmed MERS patients, there continues to be no evidence of sustained human-to-human transmission at the community level.

WHO bases this assessment on the evidence that:

1. The clinical picture seen in recent outbreaks appears to be similar to that observed throughout previous outbreaks: predominantly males above the age of 60 years with an underlying medical condition, such as diabetes, hypertension and renal failure, are at a higher risk of severe disease, including death; cases without comorbidities tend to present with milder disease or are asymptomatic;

2. Of the recent cases reported to WHO, eight were primary cases with indication of zoonotic exposure, 14 were primary cases without indication of zoonotic exposure, and 9 were secondary, nosocomial cases including 4 asymptomatic cases;

3. The cases recently reported to WHO have not resulted in sustained human-to-human transmission; however, nosocomial transmission to persons in close contact with these cases in healthcare settings was observed in three instances (clusters);

4. The size (≤ 7) and number (3) of healthcare-related clusters observed during the reporting period have remained small, and no household clusters were reported; and

5. While there is variation of the R_0 number in different settings, the overall R_0 of MERS-CoV remains to be < 1. The R_0 in outbreaks in contained health-care settings is higher, and this has been documented in healthcare-associated outbreaks in several

countries, such as Saudi Arabia and the Republic of Korea. However, within health-care settings, early isolation of cases and implementing appropriate IPC measures, R_0 can be brought to <1.

Can additional cases of MERS-CoV infection be expected to be reported from countries in the Middle East? Can additional exported cases be expected to be reported from countries outside the Middle East?

WHO expects to see additional cases of MERS-CoV infection reported from the Middle East region and that cases will continue to be exported to other countries by individuals who might acquire the infection after exposure to dromedaries or consumption or medicinal use of camel products (for example, raw camel milk or urine), or through contact with infected individuals (for example, in health-care settings). Due to the similarity of symptoms with other respiratory diseases that are widely circulating, like influenza or COVID-19, detection and diagnosis of MERS cases may be delayed, especially in unaffected countries, and provide an opportunity for local onward human-to-human transmission.

Despite a limited number of human infections reported outside the Arabian Peninsula, recent seroprevalence studies in occupationally exposed populations, MERS-CoV specific T-cell responses, and asymptomatic infections in camel-exposed African populations indicate that there is zoonotic transmission in countries across the African continent. [5,37,38] The recent detection of Clade B viruses in camels of African origin [9,10] further highlights the risk of MERS-CoV spread from the Arabian Peninsula through camel movements and poses additional risk to other geographies, especially in light of the documented increased replication competence and more efficient viral entry of Clade B over Clade C. [11]

Until more is understood about mode of transmission and risk factors for infection, cases resulting from animal to human (zoonotic) transmission will continue to occur and will eventually lead to limited community transmission within households and possibly significant healthcare-associated outbreaks such as those seen in the Republic of Korea and Saudi Arabia. Consistent application of adequate IPC measures has been used to end transmission in previous clusters.

WHO advice for MERS

WHO continues to update information products and guidance materials in collaboration with international partners, available online. [27] The latest updates are listed in the previous section on recent WHO MERS activities and include the MERS-CoV investigation protocols [22] and the WHO MERS dashboard [12]. To streamline its work on coronaviruses, promote integration into broader respiratory disease management and build on the gains and lessons learned from the COVID-19 pandemic, WHO created a Global COVID-19 and other Coronaviruses Programme in January 2025 that consolidates the work on COVID-19, MERS and SARS.

WHO emphasizes a comprehensive and collaborative approach that involves various sectors to prevent and manage the spillover and spread of MERS-CoV. Efforts include studying spillover in different geographies, evaluating transmission in healthcare settings, and enhancing IPC measures.

Prevention and Control at the Animal-Human Interface

Collaboration between human and animal health sectors in affected countries is crucial to understand MERS-CoV transmission and spillover risks, including seasonal variation. Efforts should focus on limiting the spread of infection in animal populations through development of dromedary vaccines and improved management of infected animals/herds to reduce opportunities for amplification in camels, human exposure and spillover. To better understand the geographic extent of MERS-CoV infection in dromedary camels and spillover risk into human populations, WHO encourages and supports field-based studies at the animal/human interface in countries of the Arabian Peninsula and beyond, i.e. in Africa and South and Central Asia. To enable monitoring and evaluation of virus evolution for global, regional and national risk assessment, identified viruses should be sequenced. **WHO urges Member States and scientific institutions to timely share sequence information in publicly accessible databases so that risk assessments and advanced analyses can be performed.** WHO also highlights the importance of testing for and genomic sequencing of MERS-CoV and SARS-CoV-2 given the intense circulation of SARS-CoV-2 globally and the potential for recombination of coronaviruses (i.e. creating a novel virus that combines genetic materials from two different viruses affecting the same host cell at the same time).

WHO is working closely with FAO and WOAH to better understand MERS-CoV dynamics at the animal-human interface through a **One Health approach**. Studies in the Arabian Peninsula and in countries in West and North, as well as in the Horn of Africa indicate that people handling or working with dromedary camels are at increased risk of infection with MERS-CoV compared with people who do not have contact with camels. However, dromedary camels infected with MERS-CoV may not show any signs of infection. It is therefore not possible to know whether an animal on a farm, in a market, at a racetrack or in a slaughterhouse is shedding MERS-CoV and can potentially infect humans. Infected animals may shed MERS-CoV through nasal and eye discharge, faeces, and potentially in their milk and urine. The virus may also be found in the raw organs and meat of infected animals.

Recommendations for spillover prevention: Until more evidence is gathered about infection and shedding in animals, it would be prudent for camel farm workers, slaughterhouse workers, market workers, veterinarians and anyone else handling dromedary camels to practice good personal hygiene, including frequent hand hygiene, and avoid direct contact with excretions or raw organs and meat.

Specifically:

- Hands should be washed with soap and water and/or alcohol gel before and after every contact with an animal.
- When in close contact with camels, workers should wear facial protection where feasible; and protective clothing and boots, which should be removed after work (followed by hand hygiene) and washed daily.
- Workers should avoid exposing family members to soiled work clothing, shoes, or other items that may have come into contact with camel secretions and excretions.
- Work clothes and other items should remain at the workplace for daily washing and workers should have access to and use shower facilities at their workplaces before leaving the premises.
- Unless wearing adequate personal protective equipment, people should avoid contact with any dromedary camel that has had a recent positive MERS-CoV infection until subsequent tests have confirmed that the animal is free of infection.
- Sick dromedary camels should never be slaughtered for consumption; dead dromedary camels should be safely buried or destroyed. [39]

Until more is understood about MERS-CoV, people at high risk of developing severe disease (any person who is above 60 years, has diabetes, renal failure, chronic lung disease, or is immunocompromised), should take precautions when visiting farms or markets where dromedary camels are present or engage in any activity involving close contact with dromedary camels (especially in the Middle East and Africa). These precautions include: avoiding contact with camels; not drinking raw camel milk or camel urine; and not eating camel meat that has not been thoroughly cooked.

Prevention and Control in Community and Healthcare Settings

It has been acknowledged that MERS-CoV can spread human-to-human through direct and indirect contact with infectious respiratory particles. Health-care settings are high-risk environments for transmission. Documented outbreaks in health care settings have identified risk factors for transmission to include delayed implementation of IPC practices due to delayed recognition of infected patients, as well as inadequate ventilation – especially in areas where aerosol-generated- procedures are performed.

Human-to-human transmission and large outbreaks of MERS can be prevented with early detection, appropriate isolation, implementation of IPC measures, including paying special attention to building environment features such as ventilation, as well as appropriate clinical management. However, it is not always possible to identify patients with MERS-CoV infection early because some have mild or non-specific symptoms. For this reason, it is important that all health-care facilities establish and implement clear triage policies for rapid screening and assessment of potential MERS cases and all cases with acute respiratory symptoms. It is also important for health and care workers to apply standard precautions consistently with all patients, regardless of their diagnosis, in all work practices at all times.

WHO advises that, in documented cases of human-to-human transmission in a health-care setting, all health-care contacts (e.g., health and care workers and patients sharing space with a confirmed case), household contacts, and social contacts should be tested for MERS-CoV, regardless of whether they display symptoms. Among contacts who are at higher risk of infection are those who are in direct physical contact with the patient or the patient's biological fluids before MERS was diagnosed (e.g., treating physicians, health and care workers who performed intubation, cleaning staff). For these people, multiple specimens, including lower-respiratory specimens whenever possible, should be collected and tested for MERS-CoV within the 14-day incubation period. WHO encourages full genome sequencing be conducted routinely when clusters of cases/outbreaks in human populations occur, as well as when cases are identified in humans in contact with dromedary camels. **WHO urges Member States and scientific institutions to share sequence information in publicly accessible databases in a timely manner so that risk assessments and advanced analyses can be performed.**

Recommendations for prevention of human-to-human transmission: Enhancing IPC awareness and implementation measures remains critical to preventing the possible spread of MERS-CoV in health-care facilities.

- A risk assessment and appropriate transmission-based precautions (such as droplet precautions [40]) should be added to the standard precautions when providing care to any patient with symptoms of acute respiratory infection.
- Contact precautions and eye protection should be added when caring for probable or confirmed cases of MERS-CoV infection.
- Airborne precautions [41] should be used when performing aerosol generating procedures on any suspect or confirmed cases of MERS.
- Awareness of health and care workers, especially in affected countries, should be maintained about the risk of MERS-CoV infections and appropriate IPC, i.e. through standard operating procedures and regular trainings.
- Hospital cleaning staff should be informed of and trained to take proper precautions when cleaning and disinfecting rooms of MERS patients, including linen and waste management.

In collaboration with international partners, WHO continues to work with Ministries of Health in affected and at-risk countries to improve understanding of transmission patterns and risk factors for MERS-CoV infection in community and health-care settings and to enhance early detection and IPC measures in health-care facilities to prevent onwards transmission. A better understanding of transmission in healthcare settings, especially the exposures that result in human-to-human transmission, the potential role of asymptomatic infected health and care workers, and the role of environmental contamination, is urgently needed.

Precautions in countries outside affected regions

Health officials in countries outside of the affected regions should maintain a high level of vigilance, especially those in countries with large numbers of travellers or migrant workers returning from the Middle East. Surveillance should continue to be enhanced in these countries according to WHO guidelines, along with IPC procedures in health-care facilities. WHO continues to request that Member States report all confirmed and probable cases along with information about their exposures, testing and clinical course to inform the most effective international preparedness and response.

WHO does not advise special MERS-CoV screening at points of entry, nor does it currently recommend the application of any travel or trade restrictions.

Research Needs for MERS-CoV

WHO has developed a roadmap for the R&D of MERS-specific diagnostics, therapeutics and vaccines in 2015 [42], an updated roadmap for MERS diagnostics in 2018 [43], and a broader public health research agenda for MERS in 2018 [44]. WHO, FAO and WOAH, since 2022 also together with UNEP, have repeatedly brought together technical partners to assess and advance research needed for MERS-CoV in dromedary populations, at the human-animal interface and in human populations. The last two Global Technical Meetings on MERS-CoV and Other Emerging Zoonotic Coronaviruses in November 2021 [36] and November 2023 [16] provided the opportunity to review the MERS situation globally, the research being conducted in this field, and the impact of the COVID-19 pandemic on research efforts and ways to re-focus and integrate preparedness and response activities for MERS moving forward after the pandemic.

Research needs targeted at the animal-human interface (see also Table 4):

- Strengthen systematic mapping of MERS-CoV circulation in dromedary camels, including standardized genomic sequencing and geospatial analysis to define the viral evolutionary landscape and geographic range of infection.
- Implement longitudinal surveillance and transmission studies in camel populations to evaluate temporal and seasonal variation in viral shedding, assess risk factors for amplification, and detect evidence of re-infection and immune escape.
- Characterize precise zoonotic transmission pathways by investigating high-risk food/oral exposure routes, handling and slaughter practices, and the infectious potential of camel secretions, excretions, and raw animal products.
- Quantify the true burden of zoonotic spillover in Africa, the Middle East, and South and Central Asia by expanding serologic and T-cell–based surveys among populations with occupational or cultural exposure to camels.
- Conduct integrated camel value-chain and production-system assessments to identify critical control points for transmission using combined serological, virological, genomic, and epidemiologic studies along markets, transport routes, and abattoirs.
- Prioritize identification of intervention nodes where camel-to-camel and camel-to-human transmission can be effectively interrupted or mitigated through evidence-based biosecurity and One Health policies.
- Apply behavioral and anthropological research methodologies to describe, quantify, and understand human–camel interactions and transmission risks, and to co-design risk-mitigating interventions that are culturally acceptable and socio-economically feasible.
- Accelerate development and evaluation of MERS-CoV vaccine candidates for both camels and high-risk human populations, alongside strategies for deployment and field effectiveness assessment.

Research needs targeted at communities and healthcare settings (see also Table 4):

- Integrate routine and event- or risk-based MERS-CoV testing into existing multi-pathogen respiratory surveillance systems in affected countries to detect mild and asymptomatic infections and define the full clinical spectrum of disease circulating in the community.
- Advance development and deployment of multiplex molecular diagnostic platforms capable of simultaneously detecting MERS-CoV, SARS-CoV-2, influenza viruses, and emerging coronaviruses, and establish risk-based, scalable testing strategies for both national and global surveillance systems as needed.
- Identify and quantify risk factors for health-care associated MERS-CoV infection, particularly among health and care workers, and evaluate the effectiveness of environmental controls, ventilation systems, and other IPC practices in preventing nosocomial transmission.

- Conduct studies on relevant social and behavioural aspects, such as culture and gender roles, social norms, impact of stigma, health-seeking behavior, access to information, perceptions of risk and trust, and impacts of MERS prevention and control measures, and collect context-specific data and evidence on barriers to key prevention behaviors.

- Clarify the role of silent/asymptomatic and early-stage infections in onward transmission, including whether specific occupational or community behaviors increase the likelihood of non-human or low-symptom-burden exposure leading to infection.

- Conduct targeted clinical and epidemiological investigations to better define immune response dynamics, duration of infectiousness, and determinants of severe outcomes across different patient populations.

- Further investigate cross-protection conferred by SARS-CoV-2 infection and / or COVID-19 vaccination. Preliminary observational cohort studies suggest that individuals with prior SARS-CoV-2 exposure, while lacking cross-neutralizing antibodies, may exhibit cross-reactive antibody responses to spike S2 domain and T-cell responses to MERS-CoV.

- Accelerate research, development, implementation, and real-world evaluation of medical countermeasures, including therapeutics, rapid diagnostics, and prophylactic tools, to reduce morbidity, mortality, and transmission associated with MERS-CoV.

- Accelerate development and evaluation of MERS-CoV immunotherapeutic strategies, including monoclonal antibodies, polyclonal antibody formulations (IgG- or IgY-based), and engineered antibody derivatives with enhanced neutralizing breadth. These modalities have already demonstrated potent activity against MERS-CoV in *in vitro* systems and relevant *in vivo* challenge models yet remain underdeveloped due to fragmented efforts and inadequate translational pipelines.

Additional identified gaps are listed in Table 4.

Table 4. Outstanding MERS-CoV research gaps, why they matter and which actions to take

Research gap category	Specific gap	Why it matters (evidence need)	Priority actions / Research approach	Primary actors	Priority level*
Animal–human interface	Viral circulation in dromedary camels	Unknown geographic range outside Arabian Peninsula	Genomic sequencing, geospatial mapping	MoA, FAO, WOAH	4
	Camel transmission dynamics	Re-infection and seasonal shedding unclear	Longitudinal surveillance of herds	Vet institutes, research centres	4
	Zoonotic transmission pathways	Risk from consumer and occupational practices unquantified	Exposure risk investigations (raw camel products, excretions, slaughter, aerosols)	One Health surveillance teams	5
	Environmental persistence	Role of fomites and dust uncertain	Environmental virology and stability trials	Public health labs	4
	True burden in at-risk human populations	Under-detected spillover	Standardized serology, T-cell assays	MoH, WHO reference labs	4
	Transmission risk along camel value-chains	Unknown amplification and spillover hotspots	Movement mapping, sampling at markets, transport routes, abattoirs, etc.	Camel economy stakeholders	4
	Targeted interventions	Lack of proven spillover mitigation	Biosecurity trials, community intervention pilots	Policy makers, local authorities	5
	Socio-behavioral drivers	Cultural/human behaviors poorly understood	Anthropological and behavioral studies	Universities, social scientists	5
	Camel and human vaccines	No licensed vaccine	Field vaccine trials, immunogenicity studies, alternative licensure pathways	Industry, NIH, WHO, CEPI	5
Communities and health-care settings	Detection of mild/asymptomatic infections	True burden underestimated	Multi-pathogen respiratory surveillance	MoH, hospitals	4
	Multiplex diagnostics	Syndemic co-circulation requires integrated testing	Multiplex PCR validation and adoption, where appropriate/cost-effective	Lab networks, CoViNet	5
	Healthcare-associated infections	Nosocomial outbreaks persist	Risk factor quantification, HVAC audits	Health facilities	5
	Silent transmission dynamics	Early high viral load risk	Viral kinetics, contact tracing studies	Infectious disease units	4
	Immune correlates and duration	Unclear duration of infectiousness and protective immunity	Immunology, reinfection studies	Clinical research centres	4
	Medical countermeasures	Limited treatment options	Drug R&D, field-effectiveness evaluations	Pharma, WHO	5
	Immunotherapeutics	Potent mAbs/IgY exist but not deployed	Clinical development, regulatory pathways	Academic and commercial partners	4
Additional identified gaps	Diagnostic sensitivity	Serology variability across regions	Standardization, assay benchmarking	WHO collaborating labs, CoViNet	5
	Co-infection dynamics	Disease severity may be influenced	Pathogen interaction studies	Hospitals, surveillance centers	3
	Cross-reactive antibodies from SARS-CoV-2 infection or COVID-19 vaccination	Cross-protection may decrease disease severity	Standardized serology, T-cell assays	MoH, WHO reference labs	4
	Viral evolution and recombination	Potential for novel emergent strains	Continuous sequencing and variant monitoring	National genomic consortia, TAG-VE	5
	Alternative animal hosts	Silent reservoirs possible	Wildlife/domestic animal surveillance	One Health networks	3
	Health systems preparedness	Resource-sensitive IPC lacking	Implementation science for IPC uptake	Ministries of Health	5

*Priority levels: 5 Critical — Immediate investment needed, 4 High — Significant impact if addressed, 3 Medium — Needed to shape long-term strategy, 2 Supportive — Complements core actions.

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